

Development and validation of a three pressure analysis (TPA) GT-Power model of the CFR F1/F2 engine for estimating cylinder conditions

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Abstract

The CFR engine is the widely accepted platform to test standard Research Octane Number (RON) and Motored Octane Number (MON) for determining anti-knock characteristics of motor fuels. With increasing interest in engine downsizing, up-torquing, and alternative fuels for modern spark ignition (SI) engines, there is a need to better understand the conditions that fuels are subjected to in the CFR engine during octane rating. To take into account fuel properties, such as fuel heat of vaporization, laminar flame speed and auto-ignition chemistry; and understand their impacts on combustion knock, it is essential to estimate accurate cylinder conditions. In this study, the CFR F1/F2 engine was modeled using GT-Power with the Three Pressure Analysis (TPA) and the model was validated for different fuels and engine conditions. The finite element cylinder model was applied to better estimate heat transfer and cylinder wall temperatures of the cast iron combustion chamber (piston, cylinder, and head) of the CFR engine under continuous standard knocking operation. Uncertainty of unburned gas temperature and cylinder wall temperatures estimation was analyzed depending on uncertainty in experimentally measured and prescribed model input parameters. The model was finally used to estimate the IVC conditions (trapped mass, residual gas fraction, and temperature), as well as unburned gas temperature and cylinder wall temperatures for various fuels throughout the engine cycle.

Introduction

The combination of direct injection and downsized engine displacement has enabled improved fuel efficiency and reduced emissions of gasoline turbocharged spark ignition (SI) engines. However, higher energy density has also resulted in more aggressive cylinder thermodynamic conditions, promoting the likelihood of abnormal combustion events such as knock and low speed pre-ignition (LSPI) that puts engine components at high risk.

Fuel anti-knock ratings have become more important as a way to mitigate knock in downsized SI engines. A fuel's anti-knock quality is characterized by its octane number (ON), which is commonly measured using the standard CFR F1/F2 engine. There are two octane rating methods for automotive SI engine fuels, namely the research method and motor method (ASTM D2699 and ASTM D2700, respectively) [1, 2].

Research octane number (RON) is the test measurement most commonly reported in the sale of automotive gasolines around the world, although most countries also regulate a minimum motor octane number (MON). For that reason, the validation of the Three Pressure Analysis (TPA) model used in this study has been performed on data taken from a CFR engine operating in the RON method.

While in many ways the CFR engine architecture is not representative of modern SI engines, it is still very capable of providing increased fundamental understanding of the relationship between cylinder conditions (pressure, temperature, lambda, etc.) and a fuel's propensity for auto-ignition. Estimation of the cylinder unburned gas temperature, through the use of a 1D thermodynamic engine code such as GT-Power, is helpful to better characterize the conditions that lead to auto-ignition. This detailed information can then be used by combustion modelers for improving prediction of auto-ignition [3].

Some of the sources of uncertainty in estimating the cylinder unburned gas temperature include: trapped mass, cylinder pressure, gas properties of the air-fuel mixture, and initial temperature at intake valve closing (IVC). The total cylinder trapped mass is a function of both the fresh charge (air and fuel) entering the cylinder as well as the residual gas fraction (RGF) retained from the previous cycle. In this work, the GT-Power model was employed to estimate the residual gas fraction. Cylinder pressure was measured experimentally by two types of cylinder pressure transducers, which will be discussed later. Lambda was measured by both a wide-band lambda sensor and verified by a standard AVL i60 gaseous emissions bench. The charge temperature at IVC is often a very difficult boundary condition to estimate. In this work, the GT-Power TPA model was used to solve for IVC charge temperature by matching several other experimentally measured boundary conditions. A design of experiments uncertainty analysis was performed to test each of these input parameters to the TPA model to derive the sensitivity of unburned gas temperature from variations of these parameters. One of the major outcomes of this study is to guide cylinder unburned temperature estimation using a 1D gas dynamic and thermodynamic code, such as GT-Power.

Others have used GT-Power, or similar 1D engine codes, to estimate cylinder conditions of the CFR engine during octane rating [4-6]. In some of those cases, predictive modeling is used to estimate the propensity of fuels for autoignition, while others have matched experimental cylinder pressures, indicated mean effective pressure

(IMEP), or volumetric efficiency. A common practice to concurrently match simulated cylinder pressure, IMEP, and inducted charge mass with experimental data is often to make small offsets to the intake and exhaust port pressures. However, in this work intake and exhaust port high speed pressure data was measured directly and imposed as boundary conditions to the TPA model without adjustments. So, this reduced the number of offsetting boundary conditions that could be applied to match the experimental conditions mentioned, while arriving at a variety of possible unburned gas temperature profiles. This work will study small deviations in the remaining TPA model inputs and settings to gain a better appreciation of how they influence the calculated cylinder conditions and their uncertainty. Then the model will be verified using fuels of varied physical and chemical properties.

Methodology

Argonne CFR F1/F2 engine

The standardized single-cylinder CFR F1/F2 engine is used to measure RON and MON of motor fuels based on the respective ASTM methods [1, 2]. Table 1 shows the main geometric features of the Waukesha CFR engine at Argonne National Laboratory. A typical CFR engine setup provides basically two measurements: carburetor sight glass fuel level and knockmeter knock units (KU), giving a relative air-fuel ratio and a relative measure of combustion knock, respectively. Therefore, in order to get more detailed and absolute measurements of engine operating conditions and combustion characteristics, several instrumentation upgrades were added to the standard CFR engine, without affecting engine geometry or performance during normal octane testing. For example, in order to measure absolute air-fuel ratio (and lambda), a wide-band lambda sensor was installed in the standard ASTM exhaust line between the exhaust port and the exhaust surge tank. High-speed pressure transducers were employed to obtain crank-angle resolved intake and exhaust port pressure data. Time-averaged intake and exhaust temperatures were measured by K-type thermocouples. In addition, crank-angle resolved cylinder pressure measurements were made by an AVL spark plug pressure transducer. While it is well known that flush-mounted pressure transducers offer improved accuracy over spark plug pressure transducers, simultaneous measurements from both agreed quite well during the intake and compression strokes on the CFR engine, and that the spark plug pressure transducer still gave helpful information about the onset of knock and knock intensity [7]. More detailed information on the engine controls and instrumentation upgrades can be found in Ref. [8].

Table 1. CFR F1/F2 engine specifications.

Combustion chamber	Cast iron, flat “pancake”
Compression ratio (-)	Adjustable, 4:1 – 18:1
Bore x Stroke (mm)	82.55 x 114.3
Connecting rod (mm)	254
Intake valve	180° shroud, no rotation
Exhaust valve	No shroud, rotating
Valve overlap	Positive 5 CAD
Fuel system	Carbureted
Ignition	Capacitive discharge coil to spark

The standard RON test conditions at standard atmospheric pressure (101.325 kPa) comprise of an engine speed of 600 revolutions per minute (RPM), intake air temperature (upstream of the carburetor) of 52 °C, a critical compression ratio dependent on the fuel octane number and spark timing of -13 crank angle degrees (CAD) after top dead

center (ATDC). In reality, atmospheric pressure is typically lower than 101.325 kPa and slight increases are made in compression ratio and reductions in intake air temperature to maintain similar cylinder pressure and temperature conditions. Setting the engine compression ratio and intake air temperature to compensate for the actual atmospheric pressure, a spark timing sweep was performed from -14 CAD ATDC to +6 CAD ATDC, in increments of 1 CAD, with primary reference fuel (PRF) 100 (100% iso-octane). The objective of the spark timing sweep was to collect model validation data for combustion with standard knock and non-knocking combustion. At each spark timing, useful information was recorded pertaining to engine operating conditions and combustion characteristics using the instrumentation described above. The lambda was kept constant at 0.89, which was determined to be that of peak knock for iso-octane based on a lambda sweep test. The compression ratio was maintained at 7.55:1, for which standard knock intensity (equivalent to ~ 50 KU) was achieved under RON test condition. The knock units were measured by the standard CFR D-1 Detonation Pickup installed in the cylinder head and the Model 501-C Detonation Meter signal-conditioning unit. Moreover, the knock overpressure (ringing) was measured with the spark plug pressure transducer and calculated using the standard AVL algorithm stored in AVL IndiCom.

1-D Engine Model

A one-dimensional gas dynamics and thermodynamics code, GT-Power by Gamma Technologies, was used to estimate cylinder conditions and initial and boundary conditions for the 3-D CFD simulations [9].

TPA model

The model was developed based on the three pressure analysis (TPA) method, which uses measured high-speed intake and exhaust pressure data as boundary inputs and calculates fuel burn rate from the measured cylinder pressure. This approach requires minimum modeling of engine components with valves and ports, and minimizes uncertainties in intake/exhaust systems and combustion model, allowing more accurate estimation of cylinder conditions. Figure 1 shows the TPA model of the CFR engine developed in this study.

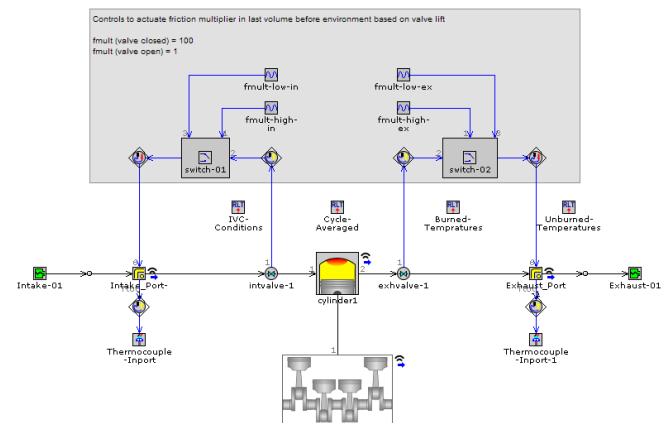


Figure 1. TPA model of the CFR engine.

At the intake and exhaust boundaries, measured instantaneous pressure, temperature, and air-fuel mixture composition are imposed. The fuel in the air-fuel mixture is assumed to be fully vaporized at the intake port inlet. In the combustion model, the measured instantaneous cylinder pressure is imposed directly to the model and

used to calculate fuel burn rate. The model works in iterative manner. The apparent burn rate is calculated before cycle start using the trapped conditions at IVC and the heat transfer of the previous cycle, and imposed to current cycle for combustion prediction. The cycles repeat until steady state convergence is reached [9]. Figure 2 shows the instantaneous pressure traces of intake, exhaust, and cylinder of PRF98 RON test, used in the model. The pressure traces were 300-cycle ensemble averaged to reduce signal noise for the burn rate calculation.

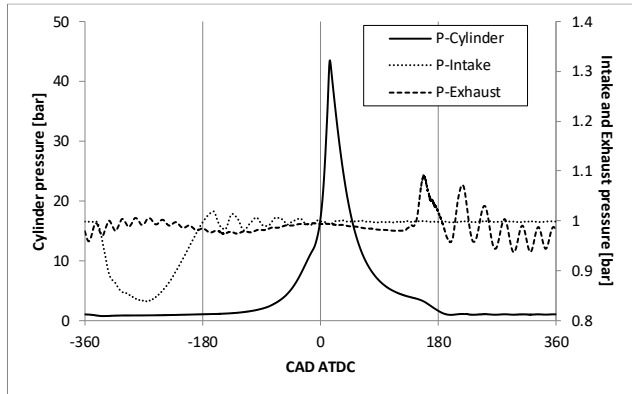


Figure 2. Cycle-averaged instantaneous cylinder, intake, and exhaust pressure traces of RON test of PRF98.

The valve specifications, such as valve timing, lift profile, and flow coefficients, are one of the important inputs to simulate volumetric efficiency accurately in the TPA model. The valve timings and lift profiles of intake and exhaust valves were measured using dial gauge at valve lash setting '0' as shown in Figure 3. The flow coefficients of intake and exhaust valves of the CFR engine were obtained from a literature and personal communication [10, 11]. The discharge coefficients were measured using a standard CFR engine cylinder head and a steady-state flow bench. The valve specifications are listed in Table 2. The valve reference diameter is basis of the flow coefficient measurement, and the valve timings are at the standard valve lash of RON/MON tests, 0.008 in.

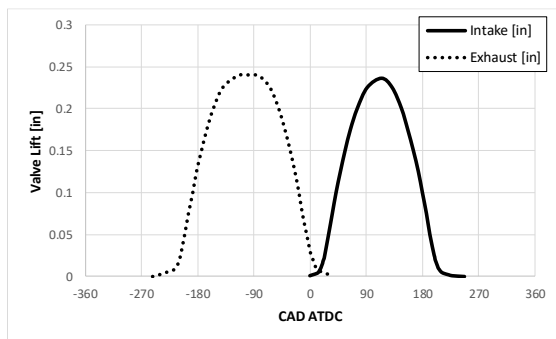


Figure 3. Measured intake and exhaust valve lift profiles at valve lash '0'.

Table 2. Valve specifications of the CFR engine.

Valve timing	IVO/IVC	376/-152
	EVO/EVC	141/373
Valve lash [in]		0.008
Valve reference diameter [in]	Intake	1.346
	Exhaust	1.356

The engine compression ratio is an important input to the TPA model in order to simulate cylinder pressure traces that match well with the

measured cylinder pressure data. Since the CFR engine is capable of a wide range of compression ratios, a correlation was derived between the cylinder head height digital counter and oil-based compression ratio measurement. In order to assess the clearance volume using oil without the risk of air bubbles in the spark plug or valve pockets, the knockmeter D-1 pickup was removed, the engine was tilted 15° with the knockmeter pickup port at the highest location (per the ASTM method), and oil was added through the knockmeter pickup port in the head until a set height [12]. With the piston at firing TDC and cylinder head height raised to the lowest compression ratio, the engine was set flat and the cylinder clearance volume recorded. The volume of oil poured into the cylinder clearance volume was assessed gravimetrically using a scale and measuring the mass loss from the beaker. From there, the cylinder head height was lowered by 50 digital counter units (higher compression ratio), excess oil above the set height in the knockmeter port was returned to the beaker and the clearance volume was reassessed. This process was repeated at intervals of 50 digital counter units from 400 to 1400. The order of the measurements was purposefully from highest head height (lowest compression ratio) to lowest cylinder head height (highest compression ratio) to not allow air bubbles entering the clearance volume from the crankcase past the piston rings. Oil-based clearance volume measurements were made across the full range stated on two separate days. For assurance that oil did not leak past the piston into the crankcase during cylinder height adjustment, these measurements were made a third time, starting in the range of compression ratios typical when rating octane number of market gasolines. Figure 4 shows the relationship found between digital counter cylinder head height and oil-based compression ratio measurement for this CFR engine. It is not clear how much this correlation may vary from engine to engine. However, it should be noted that the ASTM octane rating methods require setting the cylinder height digital counter based on a standard compression ratio and peak motoring pressure with the engine warm.

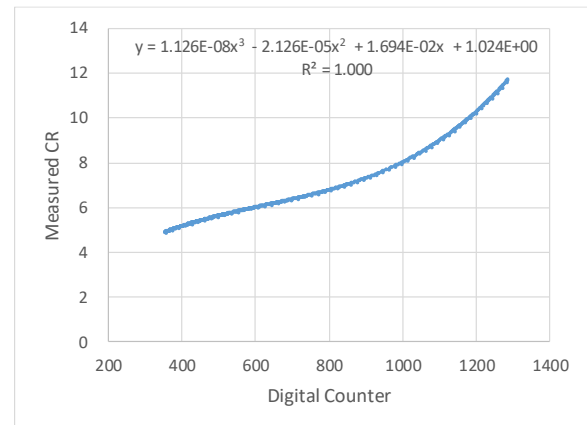


Figure 4. Measured CR of the CFR engine.

Heat transfer model

Unlike other conventional engines, the CFR engine is designed for continuous standard knocking operation, where the engine head, piston, and liner are made of cast iron to withstand extreme knock conditions. The heat transfer to the combustion chamber walls is increased during knocking cycles. It has been reported that the peak heat flux to the wall is proportional to knock intensity, and can be 2.5 times higher than for a non-knocking cycle [13]. Such conditions are very challenging to predict wall heat transfer using the typical Woschni type correlation, because the convective heat transfer coefficient might not work well for knocking conditions and it is

difficult to estimate wall temperatures at different levels of knocking operations. Considering the simple structure of the TPA model, the heat transfer is the part that causes the most of model uncertainties. To resolve these problems of the heat transfer model, the Cylinder Wall Temperature Solver (EngCyltWallSoln) and the Finite Element Cylinder Structure Geometry (FECylinderStructure) objectives in GT-Power are applied. The objectives model the cylinder structure of an engine using finite element representation of the cylinder liner, piston, and head, including ports, valves and valve guides, and predict the structure and surface temperatures. The objectives require detailed structure geometry of head, piston, cylinder, and valves as model inputs. The structure geometry of the CFR engine was obtained by 3D X-ray scan of the full engine head and cylinder assembly. Figure 5 shows the 3D CAD result of the CFR engine cylinder and head from the X-ray scan and the key geometry inputs for the “FECylinderStructure” are listed in Table 3. Heat transfer coefficients of coolant/oil cooling boundaries and structure-structure contacts were taken from the GT-Power example model and the values are listed in Table 4. The heat transfer coefficients were fixed for all cases.



Figure 5. The 3D CAD of the CFR engine cylinder and head from the X-ray scan.

Table 3. Key geometry inputs for “FECylinderStructure”

Head Material Object	Iron
Head (Deck) Thickness [mm]	12.9
Piston Top (Deck) Thickness [mm]	28.6
Piston Height [mm]	120.7
Skirt Thickness [mm]	5.0
Ring Thickness [mm]	5.0
Cylinder Wall Thickness [mm]	6.6
Cylinder Length [mm]	140
Head-Water Jacket Bottom Distance [mm]	121

Table 4. Heat transfer coefficients for “FECylinderStructure”

Cooling Boundary Conditions	
Head Coolant HTR Coefficient [W/(m ² -K)]	=5*[RPM] ^{0.8}
Piston Oil HTR Coefficient [W/(m ² -K)]	=1.5*[RPM] ^{0.8}
Cylinder Oil HTR Coefficient [W/(m ² -K)]	=1.5*[RPM] ^{0.8}
Cylinder Coolant HTR Coefficient [W/(m ² -K)]	=5*[RPM] ^{0.8}
Structure-Structure HTR	
Skirt to Cylinder HTR Coefficient [W/(m ² -K)]	5000
Ring to Piston HTR Coefficient [W/(m ² -K)]	15000
Ring to Cylinder HTR Coefficient [W/(m ² -K)]	30000
Valve to Seat HTR Coefficient [W/(m ² -K)]	50000
Valve to Guide HTR Coefficient [W/(m ² -K)]	12500
Head Gasket Contact Resistance [(m ² -K)/W]	5.00E-05
Head to Valve Seat Contact Resistance[(m ² -K)/W]	1.00E-12

Two heat transfer parameters, intake port heat transfer multiplier and combustion chamber convection multiplier, are tuned to match the model with experiment. The intake port heat transfer multiplier is used to match the volumetric efficiency and the combustion chamber convection multiplier is used to match gIMEP and cylinder pressure between simulation and experiment. The volumetric efficiency was able to be matched within $\pm 1\%$ by setting the intake port heat transfer multiplier to 4 at wide operating conditions. The combustion chamber convection multiplier is typically increased or decreased uniformly for overall cycle, but it has been found that the uniform change of the multiplier under a knocking condition in the CFR engine caused higher peak pressure during combustion and lower cylinder pressure during expansion at matched gIMEP. It is thought to be the lack of the heat transfer model capability to predict higher heat transfer during the pressure ringing due to knocking. To take this effect into account in the model, the convection multiplier is imposed as a profile, which increased the multiplier only in a few crank angle periods during the knocking, as shown in Figure 6. The combustion chamber convection multiplier was changed in range of 1 – 3 to match the gIMEP of each condition, and the Direct Optimizer function in GT-Power was used to find the target multiplier automatically.

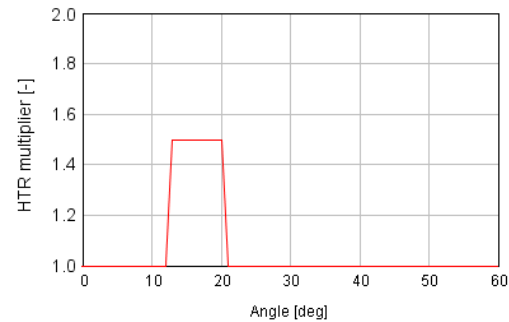


Figure 6. Profile of the combustion chamber convection multiplier for knocking condition.

Results and discussion

Model validation

The model results are validated for various fuels (PRF60-100, PRF-Ethanol blends), lambda (0.885 – 1.052), and intake port temperatures (17 - 40°C). The validation cases also include wide range of spark timing (-13 and 5 CAD ATDC) and compression ratio (5.57 to 7.34). Since the measured lambda is given in the model, the fuel flow rate is used to validate the volumetric efficiency instead. As shown in Figure 7, volumetric efficiency (fuel rate) and gIMEP from simulation showed good agreement with experiment within $\pm 1\%$. For all conditions, the intake port heat transfer multiplier was fixed to 4, and the combustion chamber convection multiplier was changed in range of 1 – 3. Measured and simulated cylinder pressure traces were compared for two spark timings, 5 and -13 CAD ATDC, under non-knocking and knocking conditions in Figure 8. The simulation agreed very nicely to experiment. The uniform combustion chamber convection multiplier was used in the non-knocking condition, while the multiplier increased only during knocking period in the knocking condition.

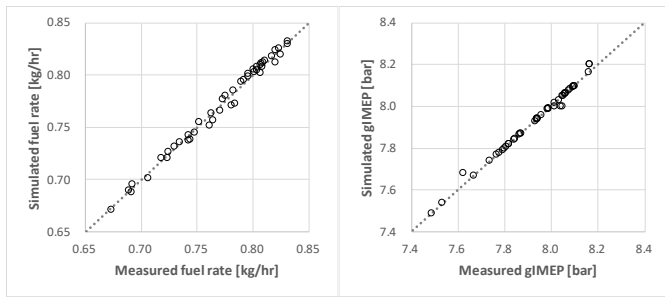


Figure 7. Model validation for volumetric efficiency (fuel rate) and gIMEP at various conditions

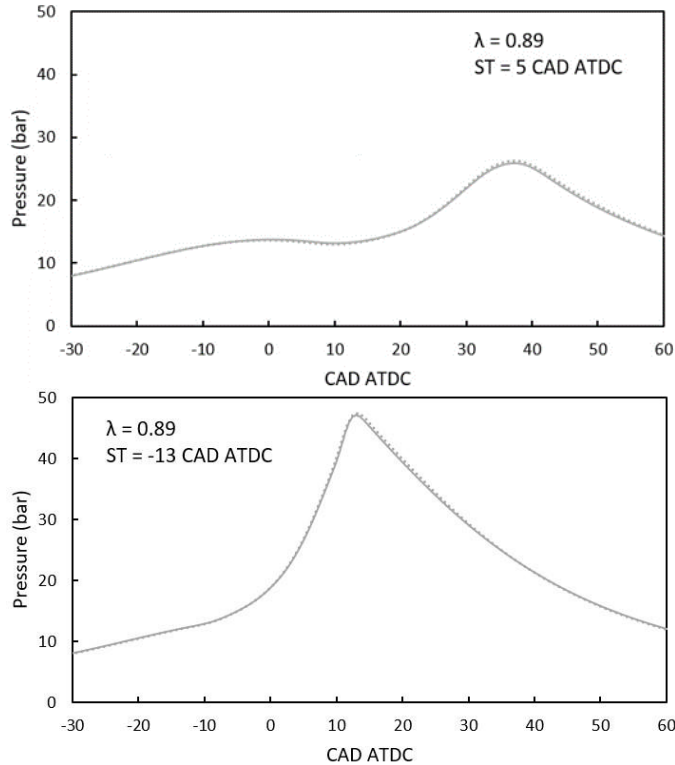


Figure 8. Measured and simulated cylinder pressure traces of PRF100 at non-knocking (ST = 5 CAD ATDC) and knocking (ST = -13 CAD ATDC) conditions

Uncertainty of the TPA model

Even though the TPA model has been validated for various CFR engine operating conditions as described in the previous section, the model has limitations: using the intake port and combustion chamber heat transfer multipliers for model matching, and relying heavily on measurement accuracy of various sensors, such as pressures, lambda, and fuel rate. This means that the model may appear to be matched even with wrong model inputs or errors in measurement by changing the multipliers. It is important to understand the uncertainty of the TPA model in predicting the boundary combustion chamber wall temperatures, IVC conditions, and unburned gas temperature.

The uncertainty of model results can occur for the following reasons:

- Unclear hardware dimensions for the Finite Element Cylinder Structure Geometry
- Unmeasurable model inputs
- Errors in measurement and sensors

To address each source of model uncertainty, following parameters were selected:

- Head, piston top, and liner thicknesses: HTR hardware
- Coolant and oil temperatures: HTR cooling
- Lambda, fuel rate, gIMEP

To find parameter combinations that matches fuel rate (VE) and gIMEP with given constraints quickly, 8-factor DoE was conducted using the AutoDoE module in GT-Power for the PRF98 standard RON condition. The eight factors and limits for DoE are listed in Table 5. The limits for the three hardware geometry factors were chosen aggressively, $\pm 50\%$ from the default value. The coolant and oil temperatures were limited to $\pm 10^\circ\text{C}$. The limit for lambda was also set quite aggressively, assuming $\pm 2\%$ error in the wide-band lambda sensor. The two HTR multipliers are included to find conditions that matches target fuel rate (VE) and gIMEP at different combinations of other factors. Total 495 DoE runs were designed using the D-Optimal Latin Hypercube for the eight factors.

Table 5. DoE factors and limits.

	Default	Min	Max
HTR Mult-Inport	4	2	6
HTR Mult-Combustion	1.5	1	3
Thickness - Head [mm]	12.9	6.45	19.35
Thickness - Piston top [mm]	28.6	14.3	42.9
Thickness - Liner [mm]	6.6	3.3	9.9
T-Coolant [K]	373	363	383
T-Oil [K]	323	313	323
Lambda	0.9047	0.8866	0.9228

After the DoE runs, established model is post processed using the Response Optimization tool in the AutoDoE module. Since the fuel rate and gIMEP are the output of the TPA model, those were set as responses in the DoE model. The errors of the fuel rate and gIMEP were also considered. The fuel rate measurement error was set to $\pm 1\%$ based on the variation of fuel rates for the repeated PRF98 standard RON tests. The error range of gIMEP was set to $\pm 1\%$ from the measured cylinder pressure traces by two pressure transducers, flush mount and spark plug types. The spark plug type transducer measured lower cylinder pressure during the combustion and expansion then the flush mount type, which caused about $\pm 1.24\%$ difference in gIMEP at knocking condition, as shown in Figure 9.

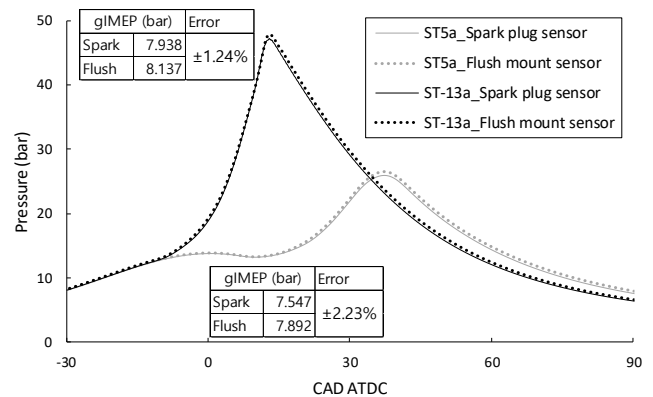


Figure 9. Measured cylinder pressure and gIMEP differences between spark plug and flush mount pressure transducers.

In the response optimization, the intake and combustion heat transfer multipliers were set as optimization independents to match the optimization targets, gIMEP and fuel rate, while the other six factors were constrained to default values or limits. Table 6 shows the cases simulated in the response optimization, which were selected to investigate separated and combined impacts of hardware factors, cooling factors, and sensor errors on uncertainty of wall temperatures, IVC conditions, and unburned gas temperature. The inport and combustion multipliers in

Table 6 were determined by the response optimization to match both the gIMEP and fuel rate within 0.1% of error. The uncertainties of wall temperatures, IVC conditions, and unburned gas temperature of each case are plotted in Figure 10. The uncertainty of each condition is described by delta from each default value. The uncertainty of RGF was low, only $\pm 0.06\%$ in maximum, and not included in the plot. The

inaccurate inputs of hardware geometry caused about $\pm 15^\circ\text{C}$ of the piston wall temperature uncertainty in maximum, but the impact for other conditions were minor. The errors of coolant and oil temperatures increased the wall temperatures uncertainties quite significantly, but did not affect much on the IVC conditions and unburned gas temperature. Among the sensors, lambda and fuel rate errors increased uncertainties for the all conditions noticeably, while the gIMEP error showed negligible impact. The sources of errors were combined to identify the maximum uncertainty of each condition. The uncertainty of wall temperatures was maximized when the heat transfer related error sources were combined. The uncertainty of piston wall temperature increased to about $+30^\circ\text{C}$ and -20°C . The maximum uncertainties of IVC conditions and unburned gas temperature were reached when all the error sources were combined.

Table 6. DoE response optimization cases. ('L': lower limit, 'H': higher limit, '-': default)

	Fixed						Opt. Target		Opt. Independent	
	Thickness (Head) [mm]	Thickness (Piston) [mm]	Thickness (Liner) [mm]	T-Cool [K]	T-Oil [K]	Lambda	gIMEP [bar]	Fuel rate [kg/hr]	HTR-Mult (Inport)	HTR-Mult (Combustion)
Default	12.90	28.60	6.60	373.0	323.0	0.9047	7.9957	0.7370	3.501	1.921
HTR (HW)	L	L	L	-	-	-	-	-	3.560	2.044
	H	H	H	-	-	-	-	-	3.595	1.958
HTR (Cool)	-	-	-	L	L	-	-	-	3.810	1.494
	-	-	-	H	H	-	-	-	2.614	1.564
Sensor (Lambda)	-	-	-	-	-	L	-	-	4.529	1.393
	-	-	-	-	-	H	-	-	3.303	2.433
Sensor (gIMEP)	-	-	-	-	-	-	L	-	3.396	2.153
	-	-	-	-	-	-	H	-	3.615	1.680
Sensor (Fuel rate)	-	-	-	-	-	-	-	L	4.119	1.588
	-	-	-	-	-	-	-	H	3.001	2.187
Combined (Max. wall T)	L	L	L	L	L	-	-	-	4.710	1.472
	H	H	H	H	H	-	-	-	3.253	1.879
Combined (Max. T-IVC)	L	L	L	L	L	H	H	H	3.127	2.218
	H	H	H	H	H	L	L	L	4.687	1.173

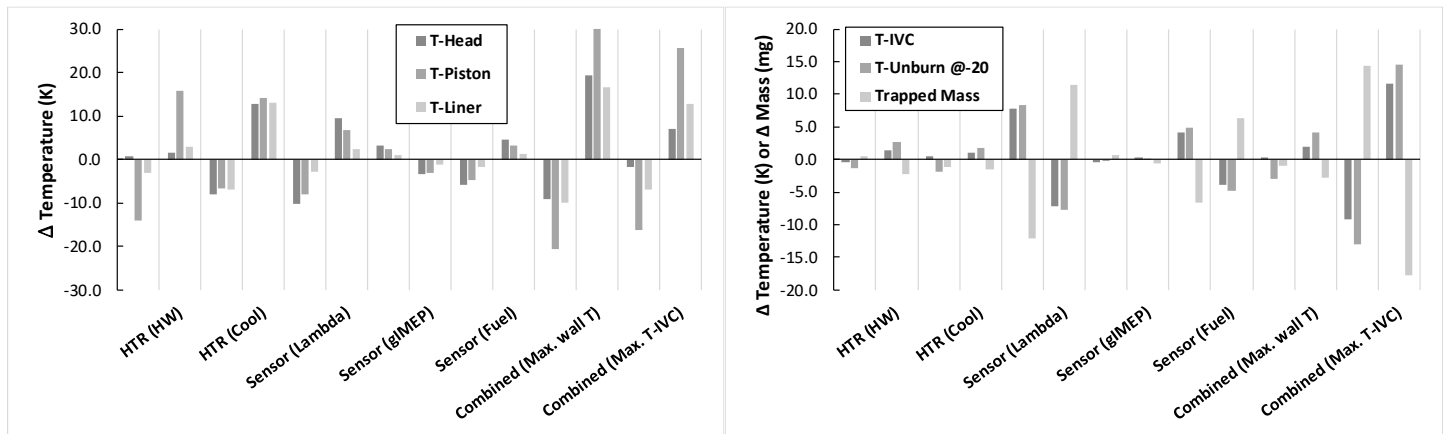


Figure 10. Uncertainty of wall temperatures (left), IVC conditions and unburned gas temperature (right) estimation of the TPA model, caused by different sources.

IVC and unburned gas temperature analysis and uncertainty tolerance for model prediction

The IVC temperature and unburned gas temperature (at 20 CAD BTDC) were estimated using the TPA model for various CFR engine test cases with three different fuels (PRF98, PRF90.2E10, PRF81.5E20, and PRF70.5E30) at various lambda and intake port temperatures. The IVC temperatures and unburned gas temperatures are plotted against the intake port temperature in Figure 11 and Figure 12, respectively. Each data series consists of different intake port temperature points at the same fuel and lambda, except for the PRF98 case, which plotted lambda 0.888-1.007 points in one data series. The IVC temperature showed a linear correlation to the intake port temperature. Interestingly, the slope of the linear regression of all data points is about 0.54, which means the temperature gap at intake port becomes about a half at IVC due to the heat transfer from the hot intake port and cylinder walls. Interestingly, the slope was kept almost constant between the same fuel and lambda cases. The IVC temperature peaks at stoichiometric fuel-air mixture and decreases as the mixture becomes leaner or richer. The wall temperatures peak at stoichiometric combustion due to the highest flame temperature, which increases heat transfer to gas during intake. The exhaust temperature peaks at stoichiometric condition, and hotter residual gas increases the temperature at IVC. However, no clear correlation was found between fuel and the IVC temperature. After compression, the gas temperatures spread wider depending on fuel and lambda, and the slope of the liner regression of all data points become about 0.66, as shown in Figure 12. The slope was also kept similar between the same fuel and lambda points. At the same intake port temperature, the temperature at 20 CAD BTDC becomes higher as the air-fuel mixture leaner (larger lambda) and more ethanol is blended, due to the increase in specific heat ratio (γ).

As discussed in the earlier section, uncertainty of the TPA model occurs through multiple sources. To ensure the reliability of the model, the results from the experimental variables, such as fuel, lambda, and intake temperature, must be sufficiently distinct compared to the uncertainty. The gray bars in Figure 11 and Figure 12 are the uncertainties of the IVC temperature and unburned gas temperatures, respectively, which are the same data sets in Figure 10. The uncertainty sourced by fuel rate and lambda measurement errors are the most significant, while the gIMEP is negligible. Since those uncertainties can be over the model resolution for experimental variables, the accuracy of the sensors should be guaranteed to get reliable results from the TPA model. Based on the result showing about $\pm 5^\circ\text{C}$ of the uncertainty with $\pm 1\%$ error of sensor (fuel rate and lambda, respectively), $\pm 0.1\%$ of accuracy of the fuel rate and lambda measurement is recommended to control the uncertainty of model estimation for the IVC and unburned gas temperatures within $\pm 0.5^\circ\text{C}$. The incorrect heat transfer parameters are another source of the uncertainty. The IVC temperature caused from the heat transfer parameters are only about 2°C in maximum, but it is equivalent to about 4°C error of the intake port temperature. Regarding the unburned gas temperature, the uncertainty can be as large as $\pm 4^\circ\text{C}$, which is greater than the temperature gap between different fuel and lambda cases. Therefore, the heat transfer related parameters, such as the geometry of finite element cylinder structure and coolant/oil temperatures are also carefully decided in the model.

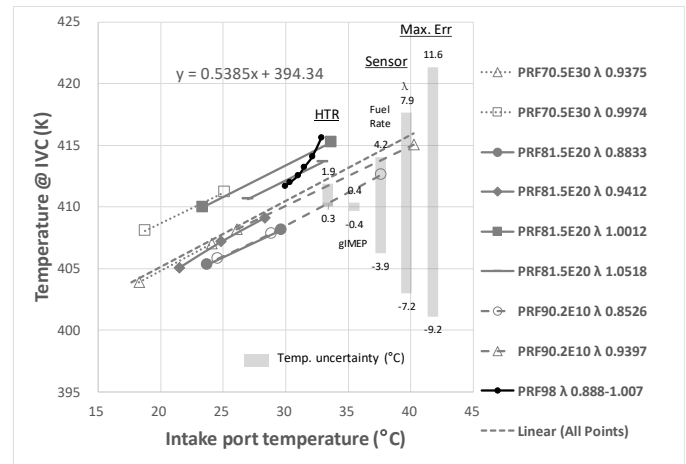


Figure 11. Intake port temperature and IVC temperature correlation, with IVC temperature uncertainty from different sources.

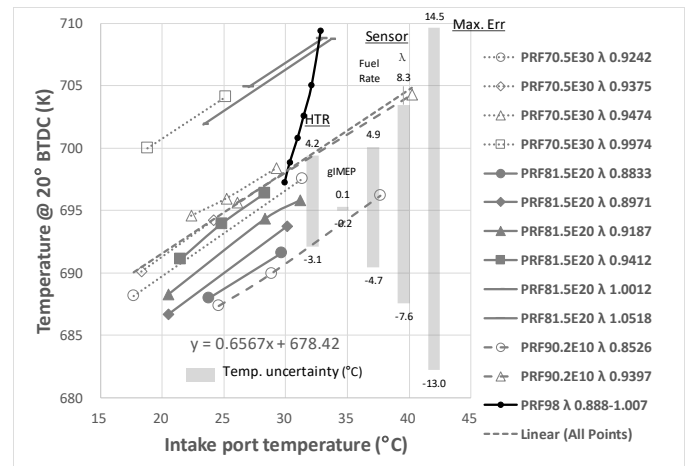


Figure 12. Intake port temperature and unburned gas temperature at 20 BTDC, with temperature uncertainty from different sources.

Summary/Conclusions

In this study, the TPA model for the CFR F1/F2 engine was developed and validated to support the experiment and 3-D simulation. The model estimates unburned gas temperature for knock analysis, and IVC conditions and wall temperatures for initial/boundary conditions of the 3-D simulation. To improve accuracy of the heat transfer model under the continuous knocking combustion of the CFR engine, the cylinder wall temperature solver with finite element cylinder structure geometry objectives were applied. The convection heat transfer multiplier is imposed as a profile, which increased the multiplier only in a few crank angle periods during the knocking. The model can be matched to the experiment by changing only the convection heat transfer multiplier in range of 1 – 3. The model is validated for wide range of the CFR engine conditions with various fuels, lambda, and intake port temperatures. The TPA model depends on experimental data and requires the heat transfer multipliers to match the model to a test result. The model is vulnerable to incorrect model input and measurement error, which causes uncertainty of the model estimation. The uncertainty of the model is analyzed quantitatively by performing the 8-factor DoE and response optimization in consideration of incorrect model inputs and measurement errors. It has been found that the uncertainty of wall temperatures is affected significantly by the incorrect inputs of heat transfer related

parameters, such as geometry of head, piston, and liner, and coolant and oil temperatures, while the impacts of measurement errors of lambda, fuel rate, and gIMEP were minor. On the other hand, the uncertainty of IVC conditions and unburned gas temperature mainly caused by measurement errors of lambda and fuel rate. The uncertainty of the IVC temperature and unburned gas temperature easily over the model resolution of different operating conditions. To ensure the reliability of the TPA model, the measurement accuracy of sensors must be guaranteed.

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